Senior AI Engineer (Ph.D. required)

Exponent · Houston, TX

Job ID (this collection)	CONSULT-108
Employer	Exponent
Location(s)	Houston, TX
Career stage	Experienced (Ph.D. plus 5+ years)
Location group	Texas (outside DFW)
Majors	Computer Science, Data Science
Consulting area	Testing, inspection & failure analysis
Pay listed	Not listed
Recruiting status when checked	Open when checked: the employer page showed the posting on Oct 28, 2026.
Employer page	https://careers-exponent.icims.com/jobs/2562/job
Captured	October 28, 2026

> Saved on October 28, 2026 from Exponent's own careers page. Everything below the line is the employer's own wording, copied word for word and checked against the live page; only the layout was changed for printing. Live listings change and close, so check the employer page before applying.

Full job posting

ID 2026-2562

Location US-TX-Houston

Practice

Computer, Data, & Security Sciences

Position Type Full-time

Workplace Type

On-Site

About Exponent

Exponentis the only premium engineering and scientific consulting firm with the depth and breadth of expertise to solve our clients' most profoundly unique, unprecedented, and urgent challenges.

Our vision is to engage multidisciplinary teams of science, engineering, and regulatory experts to empower clients with solutions that create a safer, healthier, more sustainable world. For over five decades, we've connected the lessons of past failures with tomorrow's solutions to advise clients as they innovate technologically complex products and processes, ensure the safety and health of their users, and address the challenges of sustainability.

Join our team of experts with degrees from top programs at over 500 universities and extensive experience spanning a variety of industries. At Exponent, you'll contribute to the diverse pool of ideas, talents, backgrounds, and experiences that drives our collaborative teamwork and breakthrough insights. Plus, we help you grow your career through mentoring, sponsorship, and a culture of learning. Thanks for your interest in joining our team!

Key statistics:

- 950+ Consultants
- 640+ Ph.D.s
- 90+ Disciplines
- 30+ Offices globally

Our Opportunity

We are currently seeking a Senior AI Engineer for our Computer, Data & Security Sciences Practice in Houston, TX. In this role, you will help design, build, and scale AI solutions — including generative and agentic AI systems — for client organizations across industries such as utilities, energy, life sciences, consumer electronics, and manufacturing. You will work alongside Exponent's multidisciplinary scientists and engineers to translate complex, real-world problems in safety-critical and highly regulated environments into production-grade AI applications.

You will be responsible for

- Architecting, developing, and deploying end-to-end AI solutions for clients, from data foundations through model development to production deployment
- Building generative AI applications, including retrieval-augmented generation (RAG) systems, LLM-powered assistants, and domain-specific copilots grounded in enterprise knowledge
- Designing and implementing agentic AI architectures — multi-agent workflows, tool use, orchestration frameworks, and human-in-the-loop controls — for complex analytical and operational tasks
- Developing knowledge representation layers (knowledge graphs, ontologies, semantic search) that ground AI systems in trusted, domain-specific context
- Deploying and scaling AI workloads on cloud infrastructure (AWS preferred), including containerized services, vector databases, and MLOps/LLMOps pipelines
- Rapidly prototyping and iterating on models, analyses, and visualizations in fast-paced, client-facing engagements
- Implementing responsible AI practices, including evaluation frameworks, guardrails, observability, and alignment with client governance and regulatory requirements
- Collaborating with Exponent consultants across engineering and scientific disciplines to embed domain expertise into AI solutions
- Communicating technical approaches and results clearly to both technical and executive client stakeholders
- Mentoring junior team members and contributing to the growth of Exponent's AI capabilities and reusable assets

You will have the following skills and qualifications

- Ph.D. in Computer Science, Machine Learning, Data Science, Electrical Engineering, Applied Mathematics, or a related field
- 5+ years of hands-on experience building and deploying machine learning and AI systems in production environments after the completion of the Ph.D.
- Demonstrated experience with generative AI, including LLM application development, prompt engineering, fine-tuning, and RAG architectures
- Experience with agentic frameworks and orchestration tools (e.g., LangChain, LangGraph, LlamaIndex, or comparable)
- Strong software engineering fundamentals in Python; experience with APIs, microservices, and containerization (Docker, ECS/Kubernetes)
- Hands-on experience with cloud platforms (AWS strongly preferred; Azure or GCP acceptable), including serverless and managed AI/ML services
- Familiarity with vector databases (e.g., pgvector, Pinecone, OpenSearch) and knowledge graph technologies is a plus
- Highly proficient in SQL, with the ability to work across large, complex enterprise datasets
- Experience working in consulting or client-facing roles, with the ability to manage ambiguity and deliver under tight timelines
- Excellent written and verbal communication skills
- Presently legally authorized to work in the United States; no immigration sponsorship or processing required

Preferred qualifications:

- Experience applying AI in safety-critical, regulated, or physical-infrastructure industries (utilities, energy, healthcare, manufacturing, transportation)
- Familiarity with AI governance frameworks, model risk management, and trustworthy AI principles

Applicants are encouraged to submit a CV (Curriculum Vitae) with publications (feel free to include publications that are in review or pending) [not restricted to 1 page].

Life @ Exponent

To learn more about life at Exponent and our impact, please visit the following links:

- https://www.exponent.com/careers/life-exponent
- https://www.exponent.com/company/our-impact

Attracting, inspiring, developing, and rewarding exceptional people with diverse backgrounds and expertise are central to our corporate culture. Our diverse team allows us to provide better value to our clients and enjoy an enriched work environment.

Our firm is committed to offering a variety of programs and resources to support health and well-being. We believe that providing competitive benefits as well as compensation and recognition programs empowers our staff to do work that makes a difference.

Work Environment

At Exponent, we have found that in-person interactions deepen employee engagement and are crucial for development, for realizing the full potential of our talented and diverse teams, and for building a more inclusive workplace where all have a sense of belonging. In our offices, you can expect a supportive culture and a collaborative, dynamic, multi-disciplinary work environment.Our consultants engage in-person in the office unless they are traveling for client work or other business activities.

We value the rich lives our colleagues enjoy outside of work and understand that work/life balance is critical to our employees and their well-being. Consultants have the autonomy to balance their work and personal schedules so you can meet with clients, visit inspection sites, attend conferences, and make time for priorities outside of work, too. It is this flexible, agile work style and working hours that allow our teams to drive innovation and results in their own ways, while meeting the needs of clients. #LI-Onsite

Compensation

Our consultants are rewarded for their technical and business contributions and have an opportunity to plan for future success and career growth. Exponent's total compensation plan is consistent with its expectations of the quality and quantity of work performed and with the professional standards set by Exponent. At the Associate and Senior Associate level, total compensation includes base salary, bi-weekly bonuses for high-intensity efforts, annual bonus and 401(k) employer contribution of 7% of base salary.

The base salary range for this position is dependent on experience and capabilities which will be assessed during the interview process.

Benefits you will enjoy

Access benefits information on our Life@Exponent page:

https://www.exponent.com/careers/life-exponent

Exponent is a proud equal opportunity employer. All qualified applicants will receive consideration for employment without regard to race, color, religion, sex, national origin, veteran status, disability, sexual orientation, gender identity, or any other protected status.

If you need assistance or accommodation due to a disability, you may email us at HR-Accommodations@exponent.com.

Job Locations

US-TX-Houston